EXPERIMENT - 6

Deluge, Workflows, Testing and Debugging in Zoho Creator

Library Book Issue Management

Application Name	Library Book Issue Management
Form Name	Book_Issue_Request_Form
Report Name	Book_Issue_Request_Report
Student Name	CALEB D
Student Roll No.	241401016
Experiment	6
Platform	Zoho Creator
Record Date	09 October 2026

Lab Record	Deluge validation, request initialization, availability notifications and practical testing, including defects identified, corrections applied and retest results.

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1. Objective

To automate request validation and processing using Deluge workflows, execute realistic positive and negative tests, identify defects and verify corrections.

2. Workflow Inventory

Workflow	Event	Behavior
Initialize_Book_Request	Created: Load of form	Initializes Pending status, false confirmation flag and default copies; hides staff controls on new requests.
Validate_Book_Issue_Request	Created or Edited: validation	Checks mobile, copies, date, consent and email before Available.
Set_New_Request_Status	Created: successful submission	Updates the saved record to Pending and Confirmation_Sent = false.
Notify_Available_Request	Edited: successful submission	Calls the confirmation function when status is Available and confirmation has not been sent.

3. Status Handling

Normal path: Pending > Under Review > Available > Collected.
Alternative: an unavailable request can be marked Unavailable.

Staff use report editing to set the status and internal notes. The status choices support a simple manual review process; they are not a formal Blueprint that enforces every transition.

The reusable function is **Send_Availability_Confirmation(int request_id)**. It sends through Library_Mail_Connection and updates Confirmation_Sent only after a successful API response.

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4. Deluge Validation Script

The following is the saved validation logic, printed for reference. Long lines may wrap in this record.

```
// Require a valid Indian mobile number, including country code.
phone_digits = ifnull(input.Contact_Number, "").toString().replaceAll("[^0-9]", "");
if(!phone_digits.matches("91[6-9][0-9]{9}"))
{
 alert "Enter a valid Indian mobile number: +91 followed by 10 digits starting with 6, 7, 8 or
    9.";
 cancel submit;
}
if(input.Number_Of_Copies == null || input.Number_Of_Copies < 1 || input.Number_Of_Copies > 5)
{
  alert "Number of copies must be between 1 and 5.";
  cancel submit;
}
// Reject past dates for new or rescheduled requests, but allow staff to process existing
    requests.
if(input.Preferred_Issue_Date != null && input.Preferred_Issue_Date < zoho.currentdate)
{
  unchanged_existing_date = false;
  if(input.ID != null)
  {
    existing_request = Book_Issue_Request_Form[ID == input.ID];
    if(existing_request.count() > 0)
    {
      if(existing_request.Preferred_Issue_Date == input.Preferred_Issue_Date)
      {
        unchanged_existing_date = true;
      }
    }
  }
  if(unchanged_existing_date == false)
  {
    alert "Preferred issue date cannot be in the past.";
    cancel submit;
  }
}
```

An unchanged historical issue date is allowed during processing; a newly entered or rescheduled past date is rejected.

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4. Deluge Validation Script - Continued

```
if(input.Agree_To_Library_Rules == null || !input.Agree_To_Library_Rules.contains("I agree to
    the library rules and return policy"))
{
  alert "Please agree to the library rules and return policy.";
  cancel submit;
}
if(input.Request_Status == null)
{
  input.Request_Status = "Pending";
}
if(input.Request_Status == "Available")
{
  if(input.Email_Address == null || input.Email_Address == "")
  {
    alert "Enter the member email address before marking a request Available.";
    cancel submit;
  }
}
```

5. Initialization and Notification Logic

Saved-record initialization

```
// Set_New_Request_Status: Created / Successful submission
new_request = Book_Issue_Request_Form[ID == input.ID];
new_request.Request_Status = "Pending";
new_request.Confirmation_Sent = false;
```

Availability trigger

```
if(input.Request_Status == "Available" &&
   input.Confirmation_Sent != true)
{
  thisapp.Send_Availability_Confirmation(input.ID);
}
```

The function skips example.com mock recipients. A successful send prevents another send during ordinary later edits while Confirmation_Sent remains true.

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6. Testing Procedure

1. Prepared realistic Student, Faculty and Staff requests in Development.

2. Submitted valid requests and checked persistence in the main report.

3. Attempted missing mandatory values, malformed email, invalid mobile, copy limits and past issue dates.

4. Attempted Available status without an email address and checked rejection.

5. Processed sample statuses and checked filtered reports and dashboard totals.

6. Retested each defect after the corresponding Deluge correction.

7. Test Cases and Results

Test case	Expected / observed result	Status
Empty required fields / consent	Submission rejected	Pass
Malformed email	Built-in validation rejected the value	Pass
Invalid mobile: 123	Rejected after explicit Indian-mobile validation was added	Pass
Copies: 0 or 6	Rejected; supported range is 1-5	Pass
Copies: 1 and 5	Valid boundary requests saved	Pass
New / rescheduled past date	Rejected	Pass
Available without email	Rejected with email-required alert	Pass
Staff status and notes edit	Under Review and internal notes persisted	Pass
Member-created request	Saved as Pending after initialization fix	Pass
Approved Mail API test	Delivered to owner mailbox; confirmation flag true	Pass
Later ordinary edit	No duplicate confirmation in tested flow	Pass

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8. Debugging Record

Issue	Cause / correction	Retest
Short mobile accepted	Phone input alone allowed 123. Added digit normalization and pattern 91[6-9][0-9]{9}; corrected the sample record.	Invalid phone rejected; valid +91 mobile persisted.
Blank status on member submission	Read-only member processing fields did not retain the earlier initialization. Added Set_New_Request_Status to update the saved record after creation.	Vikram saved as Pending and appeared in Staff pending queue.
Unstructured confirmation body	The previous message appeared as a paragraph. Replaced it with an HTML greeting, request-details table and collection instructions.	Approved owner-mailbox test displayed the structured email.

9. Verified Sample Data

Member	Category / copies	Status in QA snapshot
Priya Nair	Computer Science & Engineering / 2	Available
Aarav Shah	Business & Management / 5	Pending
Divya Krishnan	General Reference / 1	Under Review

Later role-test records Ananya Iyer and Vikram Rao extended the Development dataset to five. The separate Production dataset was preserved during release.

Scope of testing

These are completed test cases, not a guarantee of every possible edge case. The unchanged-old-date editing branch was implemented but was not separately exercised with an older Development record in the final QA run. Real separate-account permissions are addressed in Experiment 8.

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10 - Implementation Evidence

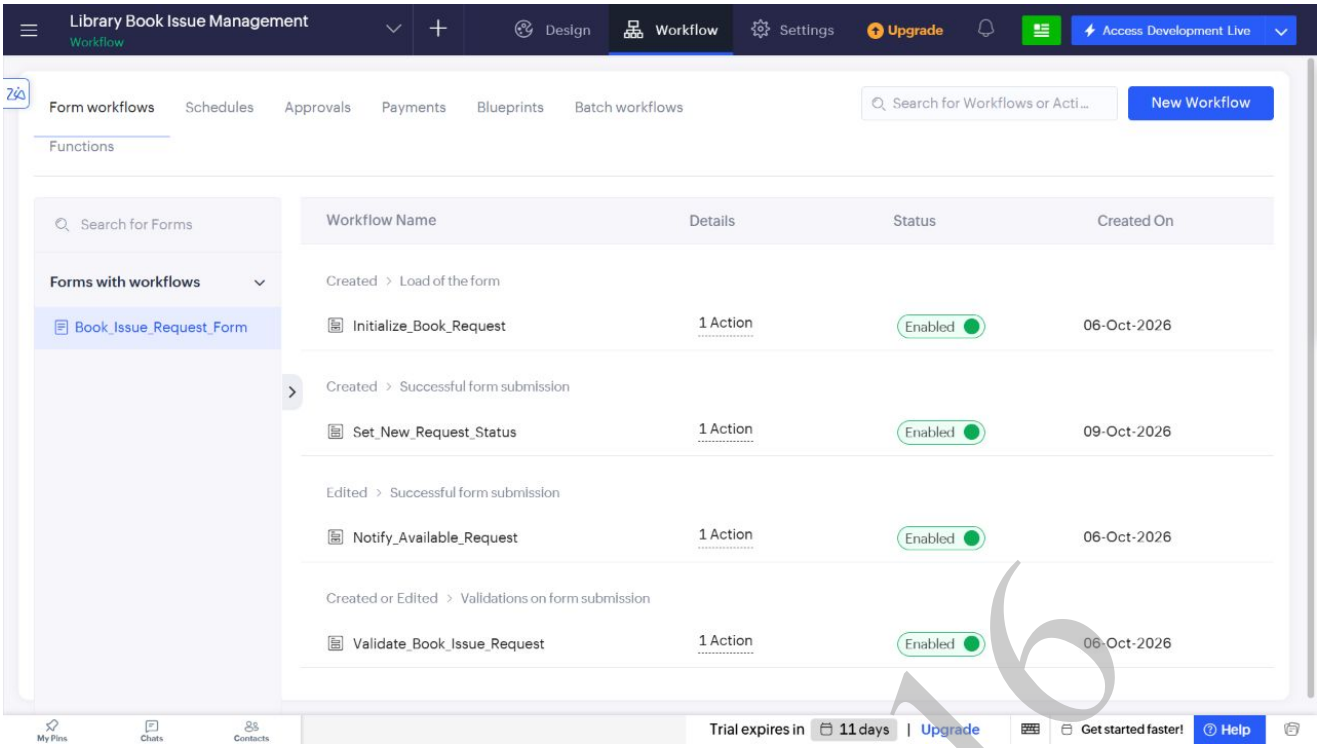


Figure 6.1. Final Development workflow list: all four named workflows are enabled.

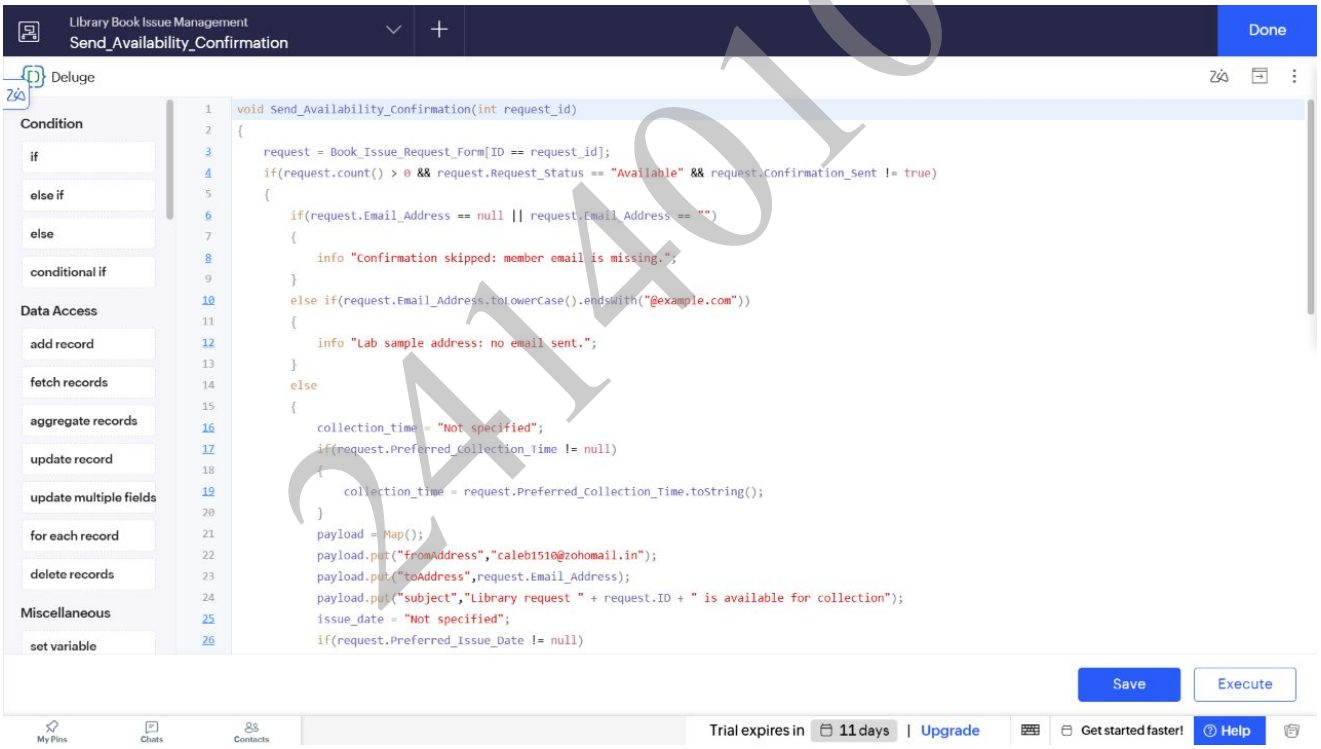


Figure 6.2. Saved Deluge source: request lookup, Available/confirmation guards, recipient checks and message construction.

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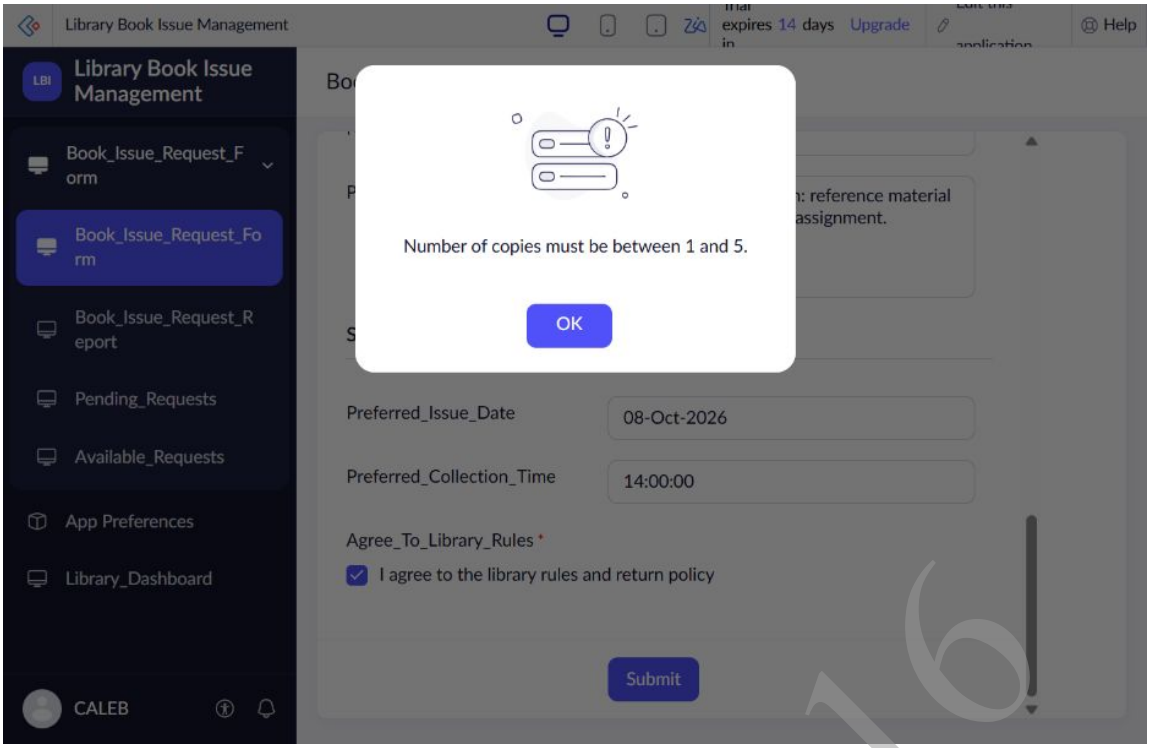


Figure 6.3. Negative test: six copies rejected with the 1-5 range validation message.

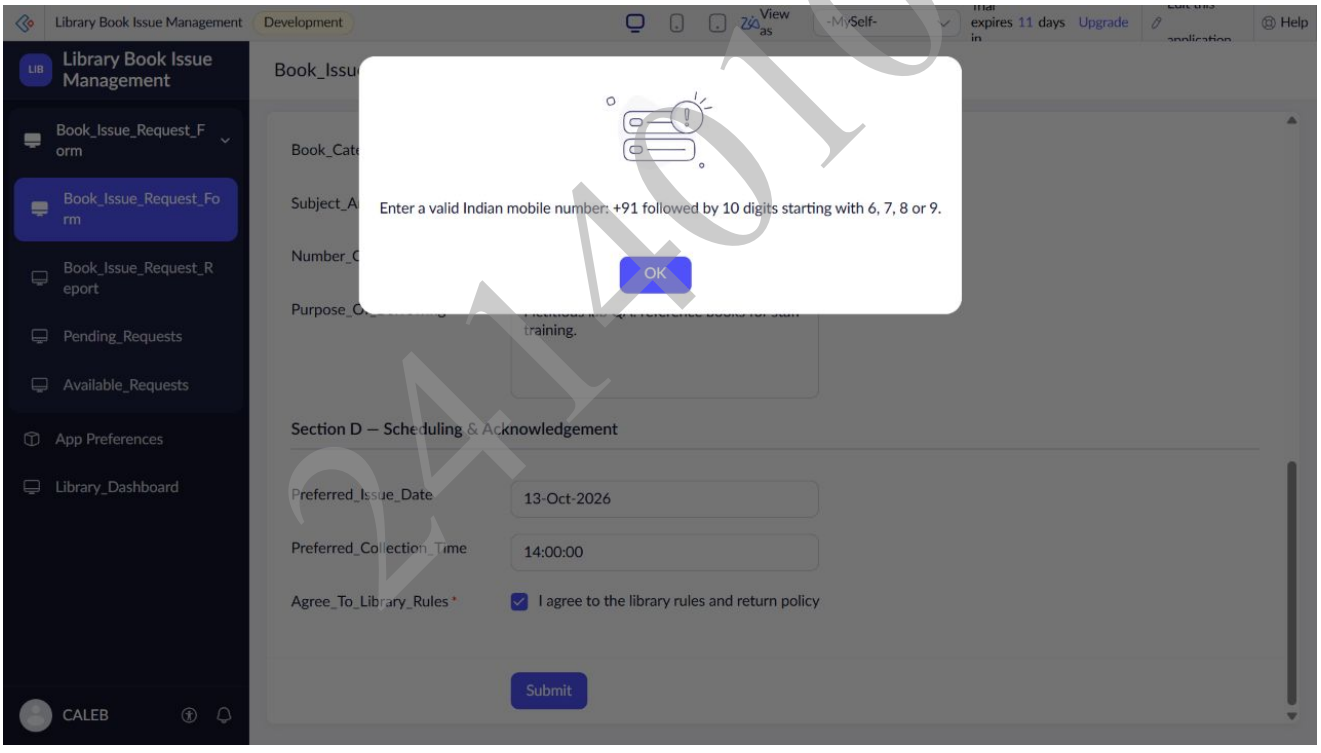


Figure 6.4. Retest after debugging: invalid Indian mobile number rejected.

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10 - Implementation Evidence

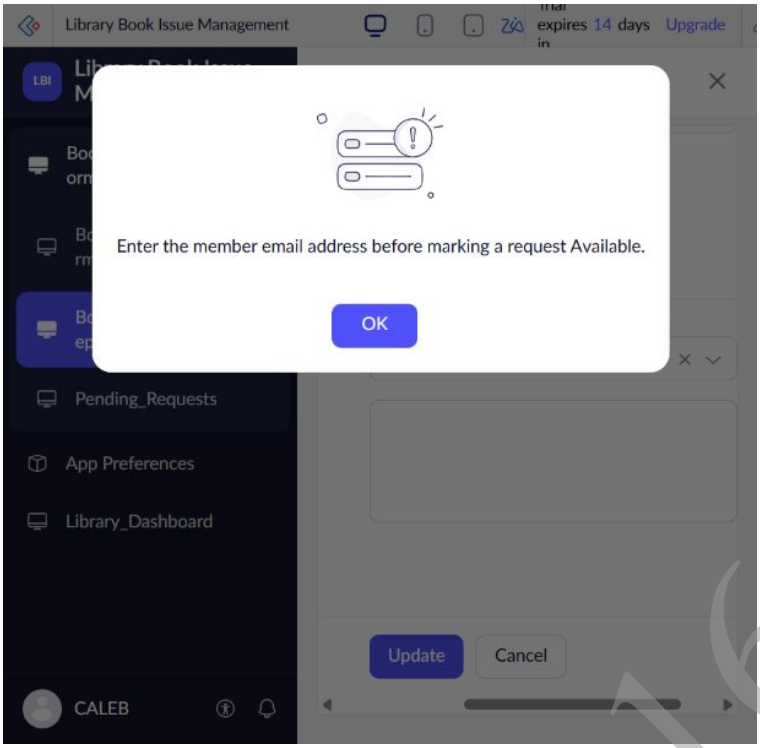


Figure 6.5. Available status blocked when the member email address is missing.

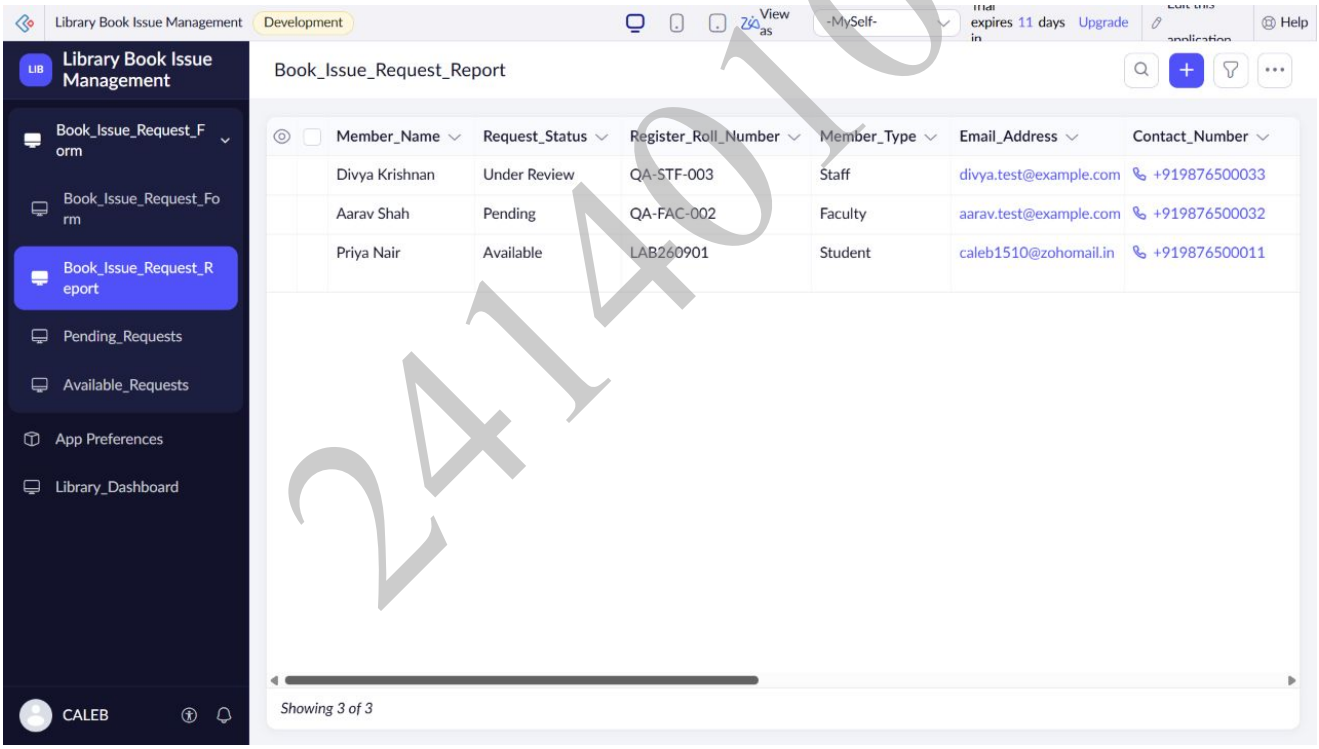


Figure 6.6. Positive tests: three realistic Development requests persisted in the report.

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10 - Implementation Evidence

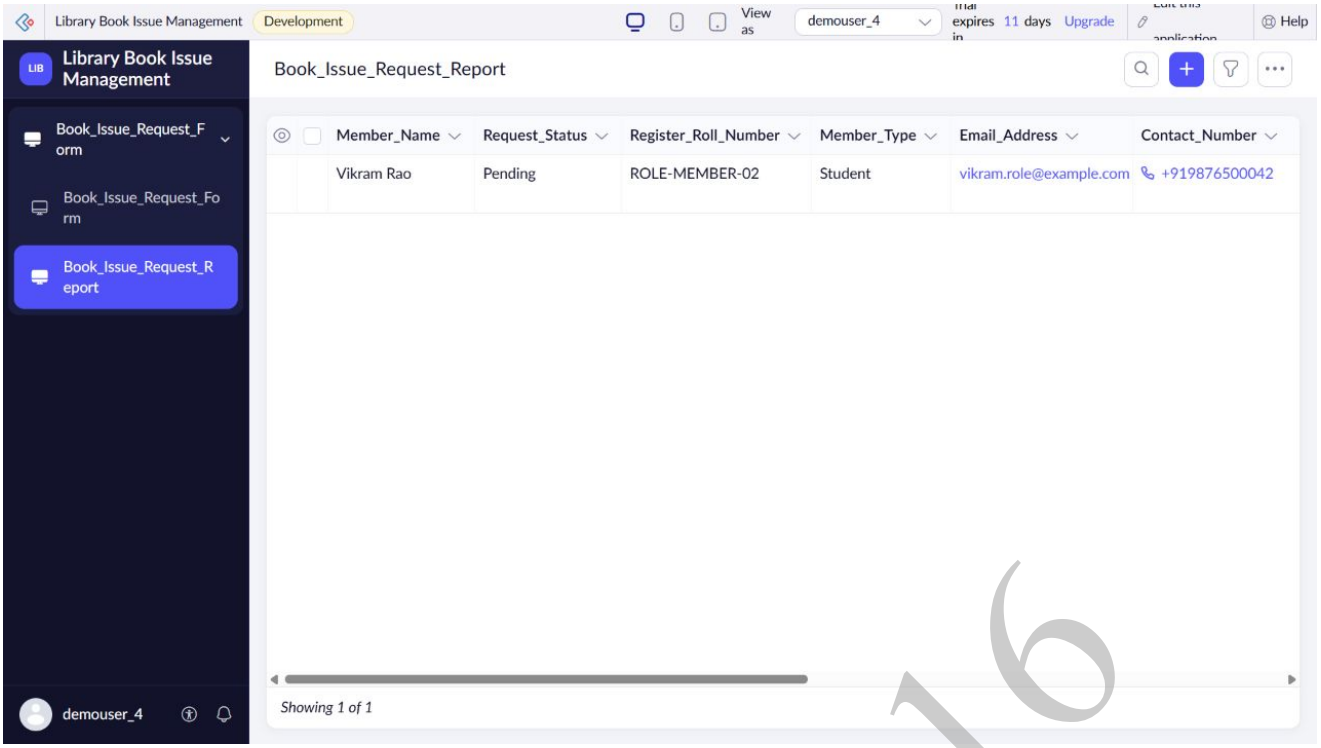


Figure 6.7. Regression test using a Development Member demo user: saved request has Pending status.

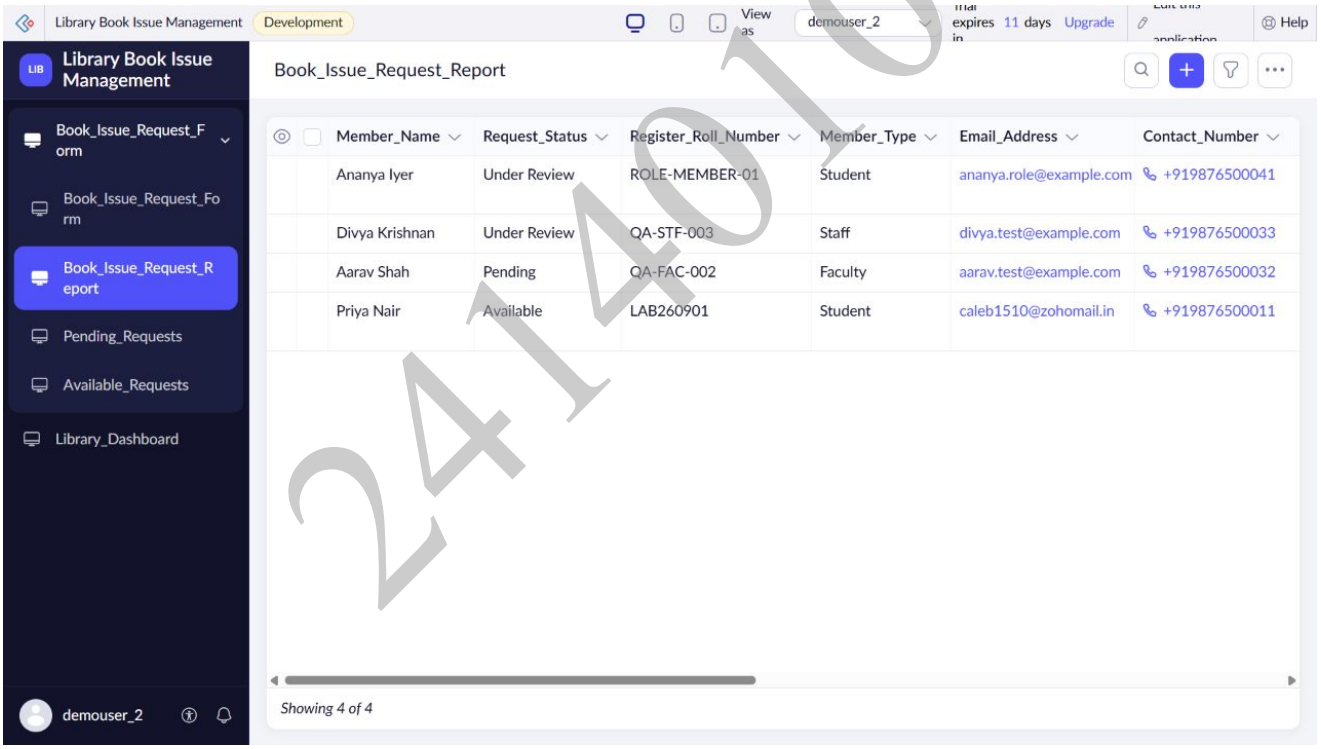


Figure 6.8. Development Staff demo-user processing evidence after the status and notes update.

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11. Expected Learning Outcomes

- Choose form events for validation, initialization and post-save actions.

- Write Deluge conditions and reusable function calls.

- Test valid, invalid and boundary inputs.

- Record defects, apply corrections and verify regression behavior.

12. Result

Four enabled workflows and one reusable Deluge function were implemented. The listed tests passed, including the mobile-validation and member-status regression tests. The final changes were included in Production v1.1.

Conclusion

Deluge workflows make the library process consistent by rejecting invalid requests, initializing saved data and triggering availability confirmation at the correct processing stage.

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